



Modern data centres deploy firewalls and managed networking components, but still feel insecure because of hackers. Hence, there is a crucial need for tools that accurately assess network vulnerability. This article brings you the top 10 assessment tools to address these issues, categorised based on their popularity, functionality and ease of use.

Vulnerabilities are unfortunately an integral part of every software and hardware system. A bug in the operating system, a loophole in a commercial product, or the misconfiguration of critical infrastructure components makes systems susceptible to attacks. Malicious techies can penetrate systems via these vulnerabilities, for personal or commercial gains. While technically this is not very easy, there have been enough successful attempts to cause one to worry. Earlier, it was believed that this was true only for commercial products. Yet, lately, open source systems have been hacked, resulting in data theft and a loss of reputation or money. Apart from local area networks, websites are also vulnerable and have become the prime target of hackers. In short, vulnerabilities can be exploited from within the organisation, as well as over the Internet by unknown people.

On the bright side, with the number of attacks increasing, there are now a slew of tools to detect and stop malware and hacking attempts. The open source world has many such utilities (and distros). Here, I must mention Backtrack Linux, which has gained international fame for its wide range of vulnerability assessment and digital forensics software utilities. The most recent version also contains powerful wireless vulnerability testing tools. Though there are literally hundreds of tools, I have selected the top 10 based on the fact that no other tool can really replace them. The primary selection criteria have been the feature set, how widespread the product is within the security community, and simplicity.

Please refer to Figure 1, which shows the top five

tools I chose for network assessment, while Figure 2 shows the leading Web vulnerability scanning products. Of course, only FOSS tools are mentioned. I have presented the tools in the order that they are expected to be used to detect vulnerabilities; this should provide a systematic approach to readers who wish to make a career as certified penetration testers.

The top five network security assessment tools

Vulnerability scanning of a network needs to be done from both within the network as well as without (from both 'sides' of the firewall). The approach I would suggest is to start from the network evaluation phase, where sniffing and primary attacks are performed. The gathered data is used in the attack phase to exploit the exposed vulnerabilities.

Wireshark: The very first step in vulnerability assessment is to have a clear picture of what is happening on the network. Wireshark (previously named Ethereal) works in promiscuous mode to capture all traffic of a TCP broadcast domain. Customised filters can be set to intercept specific traffic; for example, to capture communication between two IP addresses, or capture UDP-based DNS queries on the network. Traffic data can be dumped into a capture file, which can be reviewed later. Additional filters can also be set during the review. Typically, the tester is looking for stray IP addresses, spoofed packets, unnecessary packet drops, and suspicious packet generation from a single IP address. Wireshark gives a broad and clear picture of what is happening on the network. However, it does not have its own intelligence, and should be